

RESEARCH ARTICLE

Toward Clinical Translation of a Bimodal Fluorescence Tool for Oral Cancer Detection and Biopsy Guidance

Nemichand¹  | Bhaswati Singha Deo² | Shivam Shukla³ | Rachna Rath⁴ | Asima Pradhan^{1,2,5}

¹Department of Physics, Indian Institute of Technology Kanpur, Kanpur, India | ²Center for Lasers and Photonics, Indian Institute of Technology Kanpur, Kanpur, India | ³Department of Biomedical Engineering, Optical Imaging Laboratory, Florida International University, Miami, Florida, USA | ⁴Department of Oral Pathology and Microbiology, SCB Government Dental College and Hospital, Cuttack, India | ⁵Siksha 'O'Anusandhan (Deemed to be University), Bhubaneswar, India

Correspondence: Asima Pradhan (asima@iitk.ac.in)

Received: 26 December 2025 | **Revised:** 31 May 2026 | **Accepted:** 16 June 2026

Keywords: early diagnosis | fluorescence spectroscopy and imaging | multilayer perceptron | oral cancer | real-time diagnosis | smartphone-based device

ABSTRACT

Early- and large-scale screening and diagnosis of a disease such as oral cancer is a boon for the underdeveloped population. Our present study focuses on clinical translation of a bimodal fluorescence imaging and spectroscopy-based portable device with the potential for early and real-time diagnosis of oral cancer. This smartphone-based device collects fluorescence images and spectra from oral tissue sites within a few minutes and can assist clinicians in biopsy guidance with its advanced contrast-enhancing algorithm between normal and abnormal tissue regions. A total of 184 patients and volunteers were tested and classified using an integrated multilayer perceptron (MLP) classification scheme, in which discriminative features were extracted from both spectral and imaging data. The combined multimodal algorithm achieved strong performance, with a macro-average precision of 92.67%, sensitivity of 91.88%, *F1*-score of 91.93%, specificity of 95.89%, and overall accuracy of 94.59%.

1 | Introduction

Oral cancer is one of the most common cancers worldwide, with a particularly high incidence and mortality rate [1–3]. The high death rate is mainly due to late diagnosis, which often results from limited access to medical facilities, the unaffordable cost of diagnosis, and time-consuming diagnostic procedures [4–7]. Thus, there is a strong need for affordable, portable, and real-time diagnostic tools for early diagnosis of oral cancer [8–10]. Over time, the laboratory-based research tools with high on-field potential, which used to be bulky, expensive, and highly dependent on experts, are transforming into portable, affordable, and accessible tools to provide diagnosis facilities in each socio-economic group [11–14]. Portable diagnostic devices have become useful in addressing these challenges [15, 16]. They offer the advantage of being accessible to a larger section of the population while maintaining high sensitivity and low cost. Currently, oral cancer is diagnosed using conventional methods

such as visual inspection with white light, cytology, and biopsy, followed by histopathology [4, 17, 18]. Histopathology is considered the gold standard for cancer diagnosis, as it involves examining biopsy tissue at the molecular level to confirm the diagnosis and determine the stage of cancer progression [19]. This method requires accurate selection of the region of interest for biopsy, followed by expert analysis using specialized laboratory equipment and procedures. However, the process is time-consuming and invasive, and if the biopsy sample is not properly targeted, it may lead to inconclusive results and the need for repeated biopsies [19–21].

In cancer diagnosis, several noninvasive optical techniques are emerging as promising tools, including fluorescence imaging and spectroscopy, as well as diffuse reflectance spectroscopy [22–26]. These methods offer a valuable alternative to conventional diagnostic tools, as they are noninvasive, cost-effective, and capable of providing real-time diagnostic information

[27–29]. Among them, fluorescence imaging and spectroscopy have gained particular attention due to their ability to detect both molecular and structural changes in tissue without the need for extrinsic labeling or staining [30–34]. These techniques are also highly sensitive to capture changes in the optical properties of tissue associated with disease progression [35, 36] and are well suited to be translated into portable point-of-care diagnostic devices, ensuring both reliability and accessibility. In fluorescence spectroscopy and imaging, researchers have targeted and analyzed the alteration in the optical properties of endogenous fluorophore biomarkers such as flavin adenine dinucleotide (FAD), nicotinamide adenine dinucleotide (NADH), collagen, and porphyrins [30, 37–40]. These molecules exhibit intrinsic fluorescence that correlates with disease progression and can be detected and analyzed directly in vivo, enabling real-time assessment of tissue health. Several studies have demonstrated that the relative concentrations of FAD and porphyrins vary with disease progression [32, 41–44]. In an earlier study from our group, Kumar et al. reported a progressive increase in porphyrin fluorescence from normal to precancerous and cancerous tissues. The intensity ratio of porphyrin to FAD has been widely used as a diagnostic marker for oral cancer. Changes in the spectral profile of fluorescence signals have shown a strong correlation with different stages of the disease, indicating the potential of this technique for early detection, often before visible changes appear on the tissue surface. Fluorescence imaging further enhances diagnostic accuracy by capturing the spatial distribution of fluorescence, helping to identify regions of interest (ROIs) for more targeted spectral measurements and biopsy guidance. Selecting the correct ROI is critical for a reliable diagnosis and also helps to reduce the number of unnecessary biopsies, which remains a limitation of conventional histopathology [19, 21, 45, 46]. Several commercially available devices, such as VELscope, OralID, and ViziLite, utilize fluorescence imaging for oral cancer screening and diagnosis. These devices are advantageous in capturing a larger spatial region in a single session; however, they are limited in detecting early-stage biochemical changes that may not manifest in spatial fluorescence patterns. In contrast, fluorescence spectroscopy provides focused point illumination, yielding spectral information that reflects underlying biochemical alterations. While this approach is sensitive to molecular changes, its limited probing area restricts comprehensive screening. Therefore, integrating fluorescence spectroscopy with other imaging modalities could improve diagnostic accuracy. To the best of our knowledge, no commercially available device currently combines fluorescence spectroscopy with imaging for oral cancer screening and diagnosis, even though this technique offers higher sensitivity for detecting subtle alterations in tissue optical properties during the early stages of oral precancer. Most imaging devices use violet/blue light sources to highlight abnormal regions, where the loss of FAD fluorescence serves as a key indicator. However, the assessment of these regions largely depends on the clinician's expertise, making the interpretation subjective and positioning such devices as complementary diagnostic tools. More advanced systems, such as OralScan, integrate software-based analysis to classify risk by computing the ratio of red-to-green (R/G) fluorescence intensities. Meanwhile, spectroscopic devices are under clinical validation and hold promise for improving the accuracy and objectivity of early oral cancer diagnosis [38].

Recent studies from our group have shown that both fluorescence imaging and spectroscopy are equally effective for the detection of oral cancer [30–32]. However, a combined approach could further improve diagnostic accuracy and clinical utility. An ideal bimodal optical spectroscopy and imaging system should provide the following key capabilities:

1. Accurate selection of the ROI for precise biopsy.
2. Real-time guidance for targeted spectroscopic measurements.
3. Integrated diagnostic assessment combining both spectroscopic and imaging data.

In our previously reported work on the design, fabrication and validation of the present bimodal device, we showcased the potential of using the combination of fluorescence imaging and spectroscopy in a single system [32]. Our preclinical test results suggested that both modalities, when used together, enhance diagnostic performance and hold promise for improving early detection of oral cancer.

Beyond these advances in optical imaging and spectroscopy, artificial intelligence (AI) and machine learning (ML) have become increasingly important tools for biomedical data analysis. Unlike conventional statistical approaches, ML methods can automatically extract complex patterns, manage high-dimensional datasets, and model non-linear relationships, thereby improving classification accuracy [47]. In oral cancer research, ML has been applied effectively to both imaging and spectroscopic modalities. Convolutional neural networks (CNNs), for example, have achieved strong performance in histopathological and fluorescence imaging, while support vector machines (SVMs) and ensemble models have provided reliable discrimination of malignant versus premalignant spectra [48]. Recent studies from our group have further demonstrated how AI-driven frameworks enhance diagnostic accuracy in both smartphone-based fluorescence systems and in vivo spectroscopy platforms [30–32]. Collectively, these developments highlight the importance of integrating AI/ML with bimodal optical technologies to achieve more robust and clinically relevant diagnostic outcomes.

In this paper, we present a combined AI/ML-based analysis approach for clinical fluorescence spectroscopy and imaging data to improve diagnostic performance for oral cancer detection. Fluorescence spectra and images were acquired from the same tissue position for each patient, ensuring spatial correspondence between the two modalities. Relevant features were extracted from both spectral and imaging data and subsequently integrated within a unified AI/ML framework, where a multilayer perceptron (MLP) classifier was employed for final decision-making. The diagnostic outcomes obtained from individual modalities were compared against the combined MLP-driven analysis to quantitatively evaluate the added value of multimodal integration.

2 | Materials and Methods

We have used an in-house fabricated, 3D-printed smartphone-based bimodal device (SBBDD) for oral cancer detection, as shown

in Figure 1. The inner assembly of the device was fabricated using biocompatible, nonfluorescent medical-grade polylactic acid (PLA) for housing optical components and high-strength polyethylene terephthalate glycol (PETG) for the outer structure and support assemblies. A 405nm miniature diode laser source (Adlabs Instruments, New Delhi) is employed to illuminate the oral tissue in vivo to excite the tissue fluorophores. A Redmi K20 Pro smartphone equipped with a Sony IMX582 48 MP image sensor is used as the detector to capture the fluorescence signal from the tissue sample. The smartphone sensor features a sensor size of 1/2.0in., a pixel size of $0.8\mu\text{m} \times 0.8\mu\text{m}$, and an image resolution of 3000×4000 pixels. Additionally, the smartphone's built-in battery is utilized to efficiently power the laser source. Figure 1a presents the computer-aided design (CAD) model of the device, highlighting its major assemblies, including the mobile holder for the smartphone, cylindrical access tube for oral cavity measurements, disposable single-use tube, IM-SM switching module for modality switching, outer casing, and the device stand providing structural support to the integrated system. Detailed descriptions of these assemblies are provided in the [Supporting Information](#). The device integrates two diagnostic modalities: a fluorescence imaging module (IM) and a fluorescence spectroscopy module (SM), illustrated in Figure 1b and Figure 1c, respectively.

Switching between the imaging and spectroscopy modules is achieved using the IM-SM switching module, as shown in Figure 1a. This enables the acquisition of both fluorescence images and spectral data using a single bimodal device. The design and functionality of this device are similar to those described in our previously published work [32].

2.1 | Device Calibration and Validation

In fluorescence imaging, fluorescence data are acquired using the smartphone's built-in camera application and recorded as standard color images in JPG format. In spectroscopy, spectral images are captured in RAW image format using the Open Camera application (freely available on the Google Play Store) to preserve the spectral information without compression artifacts. The RAW spectral images are first converted to TIFF format in Adobe Photoshop for the extraction of fluorescence intensity information, and the fluorescence spectra are subsequently processed and extracted in MATLAB. In spectral images, fluorescence intensities at different wavelengths are distributed across different detector pixels. Therefore, a pixel-to-wavelength calibration was performed to establish the relationship between detector pixel positions and their corresponding wavelengths. Figure 2 shows the complete workflow of the calibration process. The details of the pixel-to-calibration process are described in the [Supporting Information](#).

Following pixel-to-wavelength calibration, the device was further validated using FAD and protoporphyrin IX (PpIX) liquid phantoms to evaluate the capability of the system to detect and quantify fluorescence signals from clinically relevant fluorophores [32]. The fluorescence spectra acquired using the developed device were compared with those obtained from a commercial spectrometer to assess spectral accuracy and detection efficiency. The detailed description of the phantom validation process and results is provided in the [Supporting Information](#).

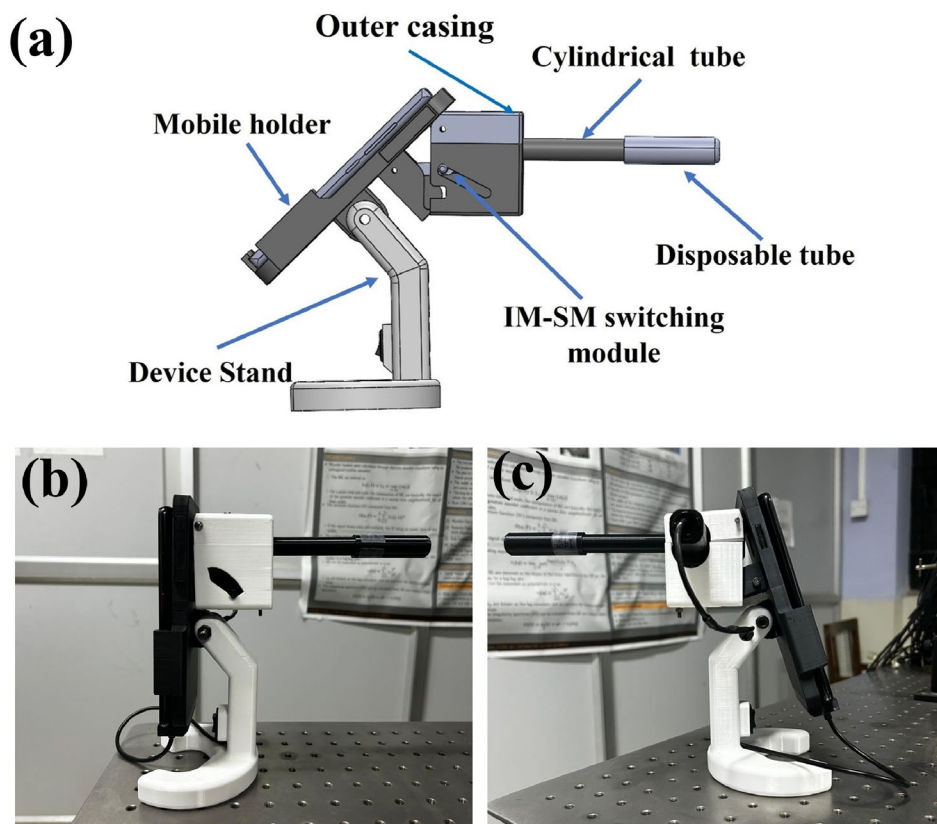


FIGURE 1 | Smartphone-based bimodal device (SBBB) developed for oral cancer diagnosis. (a) CAD model showing the major assemblies and key components of the device. (b) Imaging module configuration. (c) Spectroscopy module configuration.

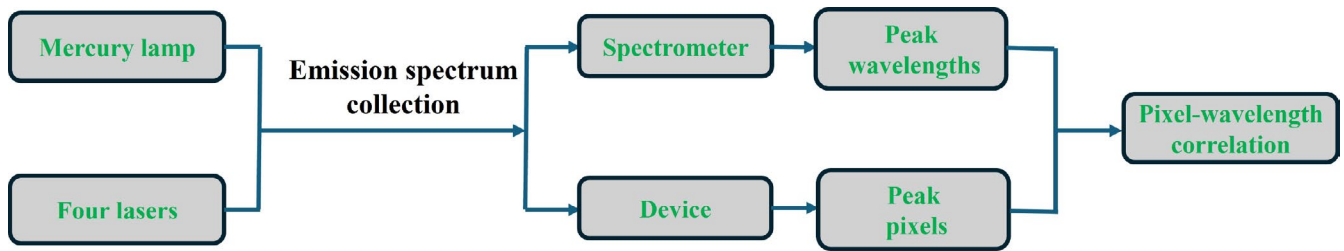


FIGURE 2 | Pixel-to-wavelength calibration workflow. The spectrometer determines the peak wavelengths of reference light sources, and the device sensor measures the corresponding pixel positions to derive the pixel–wavelength correlation.

2.2 | Data Collection Process

In this study, we employed a SBBB for in vivo testing on patients at SCB Dental College and Hospital, Cuttack (Odisha). The clinical procedures were carefully carried out under the guidance of a certified medical professional to ensure accuracy and patient safety. Figure 3 illustrates the entire testing protocol as a flowchart, with each step from patient selection to data collection. We included two groups in this study: a Control Group of healthy volunteers with no history of disease or harmful habits, and a Study Group comprising patients visiting the hospital with oral health-related issues. The Study Group participants presented with clinical symptoms such as whitish patches, reddish patches, mixed white–red patches, or raised growths. However, patients exhibiting signs of ulcers or infections were excluded following a thorough evaluation by the attending physicians. All patients visiting the hospital underwent a Clinical Oral Examination (COE) by a medical professional, which involved a light-based inspection of the clinical symptoms. Written consent was obtained from the patients in their regional language, ensuring they were fully informed about the procedures and the use of their data for research purposes. The patients were prepared for testing with the assistance of medical staff under the supervision of a medical professional. Patients visiting the hospital for treatment were instructed to refrain from consuming food or beverages and were advised to rinse their oral cavities thoroughly prior to examination. The majority of these patients reported common habits such as chewing tobacco, consuming tobacco-based pan masala, smoking, and alcohol use. The testing commenced with an IM that captured fluorescence images of the patient's oral cavities, focusing on the suspicious sites. Multiple sites were imaged to ensure comprehensive coverage of all potentially suspicious regions. These images highlighted areas with possible abnormalities. Abnormal sites were defined as clinically suspicious oral lesions identified during conventional oral examination, including leukoplakia, erythroplakia, nonhealing ulcers, mixed red–white lesions, abnormal tissue thickening, surface irregularities, and regions exhibiting clinically visible dysplastic or malignant characteristics. Once these regions were identified through imaging, the fluorescence SM was used to capture spectral images at the detected abnormal sites in the oral cavity. The abnormal sites identified by the IM were subsequently confirmed by the attending medical professional. Biopsy samples were acquired from the suspicious sites identified by the IM and further confirmed by the medical professional. The fluorescence spectroscopy and imaging findings were cross-referenced with histopathology reports to validate the

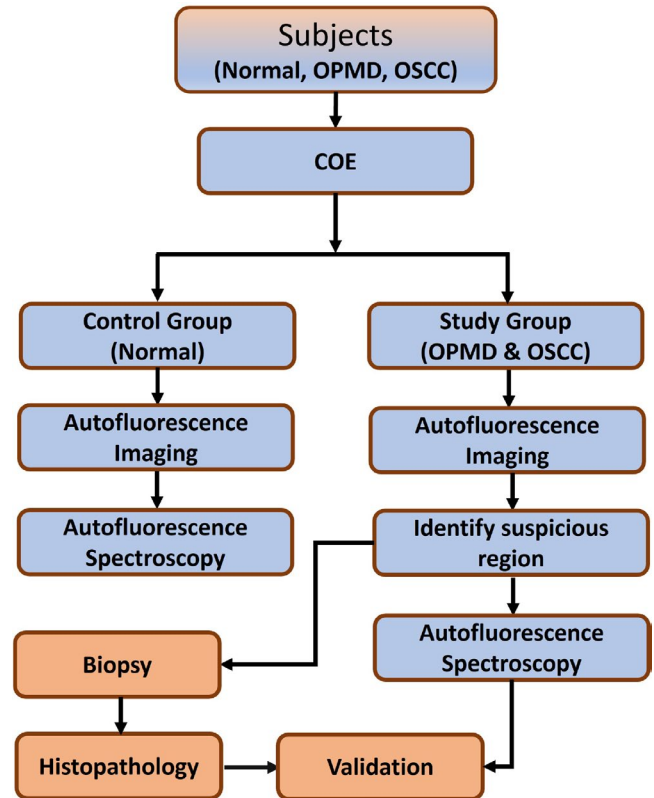


FIGURE 3 | Flowchart of the clinical in vivo testing procedure using the smartphone-based bimodal device (SBBB).

accuracy of the fluorescence-based diagnosis using the SBBB. Comprehensive details of the image conversion, calibration, and validation workflow are explained briefly in Section 2.1 and also available in our earlier publications [32, 49]. The data collected from patients were further used for analysis and classification to differentiate between normal and abnormal clinical cases.

2.3 | Data Statistic

The study involved a total of 184 subjects, distributed as follows: 59 normal, 66 precancer, and 59 cancer. Each subject underwent testing with a SBBB equipped for both imaging and spectroscopy, ensuring measurements were obtained from the same anatomical location. For spectroscopy, a single spectrum was recorded per patient, whereas for imaging, an average of four images was acquired per patient. The complete dataset statistics are summarized in Table 1.

TABLE 1 | Summary of dataset statistics used in this study.

Class	Subjects	Spectra	Images
Normal	59	59	266
Precancer	66	66	185
Cancer	59	59	194
Total	184	184	645

2.4 | Analysis Methodology

The proposed classification framework leverages multimodal inputs, specifically (i) oral tissue images and (ii) their corresponding spectral measurements as shown in Figure 4. Each modality undergoes a dedicated preprocessing pipeline, followed by feature standardization, and the final predictions are obtained through late fusion.

2.4.1 | Image Feature Extraction

Each subject contributed between 2 and 7 images. All images were first resized to 224×224 pixels and normalized to the range $[0, 1]$. Grayscale representations, denoted as $I \in \mathbb{R}^{224 \times 224}$, were then obtained by averaging the three RGB channels. From these grayscale images, a comprehensive set of descriptors was extracted to characterize textural, statistical, and structural properties, as described below.

1. Discrete wavelet transform (DWT) statistics: A single-level 2D DWT using the Daubechies-1 (db1) wavelet was applied to each image. This decomposition produces one approximation subband (c_A) and three detail subbands (c_H, c_V, c_D), corresponding to horizontal, vertical, and diagonal frequency components, respectively.

For each subband $c \in \{c_A, c_H, c_V, c_D\}$, we extracted two statistical descriptors: the mean and standard deviation, defined as

$$\mu_c = \frac{1}{N} \sum_{i=1}^N c_i, \quad \sigma_c = \sqrt{\frac{1}{N} \sum_{i=1}^N (c_i - \mu_c)^2},$$

where N is the number of coefficients in subband c .

Thus, the DWT analysis yields a total of eight features:

$$\{\mu_A, \sigma_A, \mu_H, \sigma_H, \mu_V, \sigma_V, \mu_D, \sigma_D\},$$

which collectively characterize both the global energy distribution and the variability of localized structures across different frequency subbands.

2. GLCM texture features: The gray-level co-occurrence matrix (GLCM) was computed from I after rescaling intensities to 8-bit range (0–255). Parameters included: pixel offset distance $d = 1$, angle $\theta = 0^\circ$, levels = 256, with symmetry and normalization enabled.

From the GLCM $P(i, j)$, six Haralick features were extracted:

$$\begin{aligned} \text{Contrast} &= \sum_{i,j} (i-j)^2 P(i, j), \\ \text{Dissimilarity} &= \sum_{i,j} |i-j| P(i, j), \\ \text{Homogeneity} &= \sum_{i,j} \frac{P(i, j)}{1 + |i-j|}, \\ \text{Energy} &= \sum_{i,j} P(i, j)^2, \\ \text{Correlation} &= \frac{\sum_{i,j} (i - \mu_i)(j - \mu_j) P(i, j)}{\sigma_i \sigma_j}, \\ \text{ASM} &= \sum_{i,j} P(i, j)^2, \end{aligned}$$

where μ_i, μ_j, σ_i , and σ_j are marginal means and standard deviations. These features capture second-order statistics of intensity variations, reflecting regularity, smoothness, and texture complexity in tissue images.

3. Statistical moments of intensity histogram: To capture global intensity distributions, we computed four statistical moments of the grayscale histogram:

$$\begin{aligned} \mu_I &= \frac{1}{N} \sum_{i=1}^N I_i, \quad \sigma_I = \sqrt{\frac{1}{N} \sum_{i=1}^N (I_i - \mu_I)^2}, \\ \text{Skewness} &= \frac{\frac{1}{N} \sum_{i=1}^N (I_i - \mu_I)^3}{\sigma_I^3}, \quad \text{Kurtosis} = \frac{\frac{1}{N} \sum_{i=1}^N (I_i - \mu_I)^4}{\sigma_I^4}. \end{aligned}$$

These descriptors quantify brightness, contrast, asymmetry, and peakedness of the image intensity distribution.

4. Shannon entropy: The global randomness of pixel intensities was measured using Shannon entropy:

$$H(I) = - \sum_{k=1}^L p_k \log_2(p_k),$$

where L is the number of intensity levels and p_k is the normalized frequency of level k . Entropy provides a measure of complexity, with higher values indicating greater irregularity and information content.

5. Sobel edge statistics: To characterize structural gradients, the Sobel operator was applied to compute the edge magnitude map E . The mean and standard deviation of E were extracted:

$$\mu_E = \frac{1}{N} \sum_{i=1}^N E_i, \quad \sigma_E = \sqrt{\frac{1}{N} \sum_{i=1}^N (E_i - \mu_E)^2}.$$

These descriptors capture both the overall edge density and variability of edge responses, reflecting boundary sharpness and tissue irregularities.

Each image was ultimately represented by a 21-dimensional feature vector. For subjects with multiple images, per-feature averages were computed across all images, resulting in a single 21-D image-level descriptor per subject.

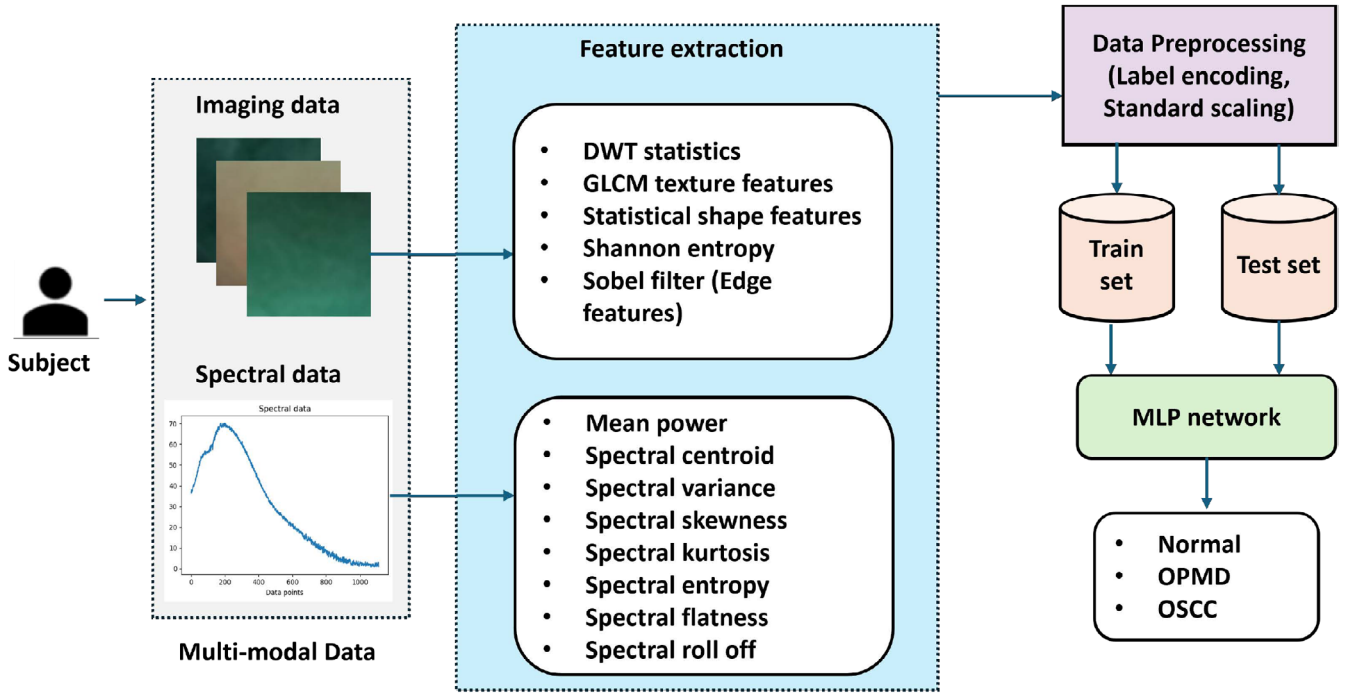


FIGURE 4 | Workflow of the proposed methodology.

2.4.2 | Spectral Feature Extraction

The spectral modality consisted of measurements represented as vectors of 1112 data points, corresponding to intensity values sampled uniformly across the wavelength range from x to y nm. Each spectral vector $\mathbf{x} \in \mathbb{R}^{1112}$ was analyzed using a one-dimensional fast Fourier transform (FFT) to capture frequency-domain characteristics. From the resulting power spectrum, we derived a set of eight Fourier-based descriptors that quantify the distribution, variability, and complexity of spectral energy.

Let $X(f)$ denote the Fourier transform of $\mathbf{x}$ and $P(f) = |X(f)|^2$ its power spectrum, computed only for non-negative frequency bins. The total spectral energy is given by

$$E = \sum_{f \geq 0} P(f).$$

From $P(f)$, the following descriptors were extracted:

1. Mean spectral power:

$$\mu_P = \frac{1}{N} \sum_{f \geq 0} P(f),$$

reflecting the average distribution of power across frequencies

2. Spectral centroid:

$$C = \frac{\sum_{f \geq 0} f \cdot P(f)}{\sum_{f \geq 0} P(f)},$$

representing the “center of mass” of the spectrum and often perceived as a measure of spectral brightness.

3. Spectral variance:

$$\sigma_P^2 = \sum_{f \geq 0} (f - C)^2 \cdot \frac{P(f)}{E},$$

quantifying the spread of frequencies around the centroid.

4. Spectral skewness and kurtosis: Higher-order moments of the spectral distribution, defined as

$$\text{Skewness} = \frac{\sum_{f \geq 0} (f - C)^3 \cdot P(f)}{E \cdot \sigma_P^3}, \quad \text{Kurtosis} = \frac{\sum_{f \geq 0} (f - C)^4 \cdot P(f)}{E \cdot \sigma_P^4}.$$

These capture asymmetry and peakedness of the spectral distribution.

5. Spectral entropy:

$$H_s = - \sum_{f \geq 0} p(f) \log(p(f) + \epsilon), \quad p(f) = \frac{P(f)}{E},$$

where ϵ prevents numerical instability. Entropy measures the degree of disorder, with higher values indicating more uniform energy spread.

6. Spectral flatness:

$$F = \frac{\exp\left(\frac{1}{N} \sum_{f \geq 0} \log(P(f) + \epsilon)\right)}{\frac{1}{N} \sum_{f \geq 0} P(f)},$$

quantifying whether the spectrum is noise-like (flat) or tone-like (peaked).

7. Spectral roll-off (85%): The roll-off frequency f_r is defined such that

$$\sum_{f=0}^{f_r} P(f) = 0.85E,$$

that is, the frequency below which 85% of the total spectral energy is concentrated.

Together, these measures form an eight-dimensional spectral feature vector for each subject. They collectively capture the energy distribution, shape, smoothness, and concentration of the spectral profile, providing a compact yet discriminative representation of the spectroscopy data.

2.4.3 | Feature Standardization and Fusion

The image descriptors (21 features) and spectral descriptors (8 features) were standardized independently using z-score normalization:

$$z_i = \frac{x_i - \mu}{\sigma},$$

where μ and σ denote the mean and standard deviation of each feature, computed across the dataset.

Following normalization, the two feature sets were concatenated to construct a 29-dimensional multimodal feature vector for each subject:

$$\mathbf{z} = [\mathbf{z}_{\text{image}} \parallel \mathbf{z}_{\text{spectral}}],$$

where $\parallel$ denotes vector concatenation.

2.4.4 | Model Architecture

The multimodal feature vectors of dimension 29 (21 image-based and 8 spectral features) were classified using a compact MLP. The dataset was divided into training and testing subsets in an 80:20 ratio, resulting in 147 training samples and 37 test samples. The network comprised an input layer followed by two fully connected hidden layers with 64 and 32 neurons, respectively, each activated by the ReLU function. To mitigate

overfitting on the relatively small dataset, both layers incorporated ℓ_2 weight regularization ($\lambda = 10^{-3}$) together with dropout at a rate of 0.4. Finally, a softmax output layer produced class probabilities across the diagnostic categories. This architecture provided a balance between expressive power and regularization, enabling the network to capture non-linear relationships among features while maintaining generalization ability.

3 | Results and Discussion

Fluorescence spectra and their corresponding images for three diagnostic categories—normal, oral potentially malignant disorders (OPMDs), and oral squamous cell carcinoma (OSCC)—are shown in Figure 5. The fluorescence images and the extracted three-band images for the same samples (shown in Figure 5) are presented in Figure 6. A clear trend is observed across these categories: the green band intensity decreases from Normal to OSCC, while the red band intensity increases. This pattern indicates an increasing contribution of porphyrin fluorescence with disease progression.

Figure 5 displays the normalized fluorescence spectrum and image for Normal, OPMD, and OSCC tissue sites. In normal tissue, the spectrum does not exhibit a significant porphyrin emission peak. In contrast, both OPMD and OSCC tissues show a prominent peak in the 630–640 nm range, with the intensity being higher in OSCC than in OPMD. This spectral feature is consistent with the known fluorescence emission of porphyrins and suggests increased porphyrin accumulation in malignant tissue. These findings demonstrate that both imaging and spectroscopy modalities capture complementary aspects of disease-related biochemical changes. The increase in red-band intensity in fluorescence images aligns with the emergence of porphyrin peaks in the spectra, reinforcing the diagnostic relevance of combining both data types. Integrating features from spectroscopy and imaging can therefore enhance classification accuracy and improve the reliability of oral cancer diagnosis.

3.1 | Data Analysis

This section presents the results of applying the proposed framework to classify normal, OPMD, and OSCC oral cancer classes using fluorescence spectral and imaging data. An 80–20 train–test split was applied to the dataset. The network was optimized

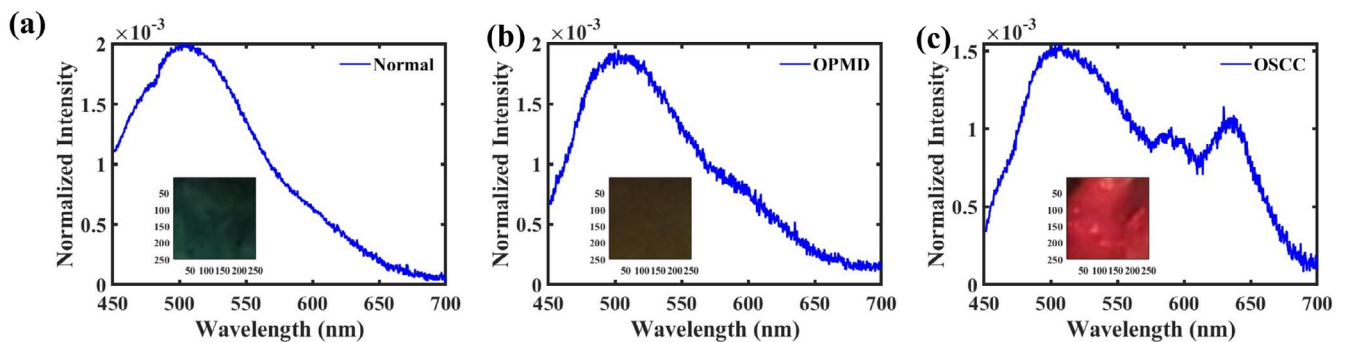


FIGURE 5 | Area-normalized typical fluorescence spectra and corresponding fluorescence images collected from the same anatomical site for: (a) normal, (b) OPMD, and (c) OSCC tissue.

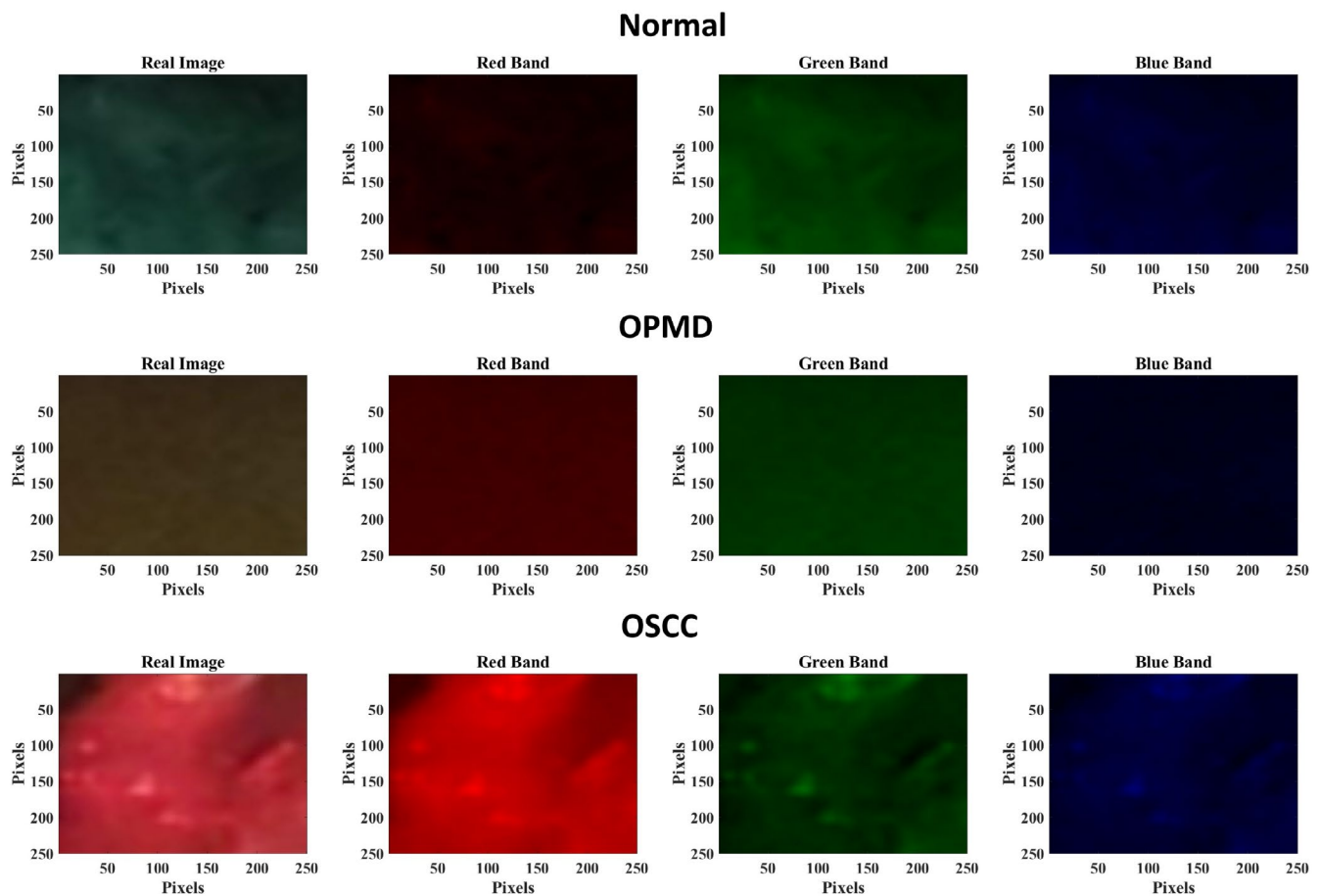


FIGURE 6 | Fluorescence images of the anatomical site and corresponding color band images highlighting spatial fluorescence patterns for normal, OPMD, and OSCC classes.

using the Adam optimizer with categorical cross-entropy loss and accuracy as the evaluation metric. Training was performed for up to 200 epochs with a batch size of 16, and early stopping (patience of 10 epochs) was employed to prevent overfitting. In addition, a model checkpointing strategy was used to retain the version of the model that achieved the highest validation accuracy. This configuration ensured robust convergence and reliable generalization performance despite the relatively small dataset size.

The classification performance of the proposed framework was evaluated across three experimental setups: using only images, only spectra, and the combined multimodal features (images + spectra). In the unimodal settings, spectral features derived from spectra and image features extracted from imaging data were independently used for classification. In the multimodal setting, both feature types were integrated using feature-level fusion by concatenating the extracted image and spectral feature vectors into a unified representation and classified through the same pipeline, thereby enabling a rigorous comparison between unimodal and multimodal approaches for distinguishing normal, OPMD, and OSCC cases. The corresponding confusion matrices on the test data are shown in Figure 7, and the detailed performance metrics on the test set are summarized in Tables 2–4.

When only image features were used, the model achieved an overall macro-average precision of 81.96%, sensitivity of 81.68%,

$F1$ -score of 81.76%, specificity of 91.60%, and accuracy of 88.63%. The spectral modality alone resulted in comparable performance, with macro-average precision of 82.05%, sensitivity of 81.20%, $F1$ -score of 81.53%, specificity of 90.50%, and accuracy of 87.39%.

The fusion of image and spectral descriptors notably enhanced the classification performance. The combined model achieved the highest macro-average scores, with precision of 92.67%, sensitivity of 91.88%, $F1$ -score of 91.93%, specificity of 95.89%, and accuracy of 94.59%. These results highlight the complementary nature of imaging and spectral modalities, where the integration of structural and biochemical information enables more accurate discrimination of normal, OPMD, and OSCC classes. The robustness of the multimodal framework was further validated through receiver operating characteristic (ROC) analysis. As shown in Figure 8, the model exhibited excellent separability across all three classes, yielding area under the curve (AUC) values of 0.99 for normal, 0.98 for OPMD, and 0.98 for OSCC.

The integrated analysis of clinical spectroscopy and imaging outcomes significantly enhances diagnostic accuracy compared with single-modality systems. The IM is not only used for fluorescence imaging but also guides spectroscopy measurements, thereby overcoming the limitation of probing only small regions. Suspicious ROIs are identified by R/G imaging and targeted for spectroscopy measurements, ensuring biochemical

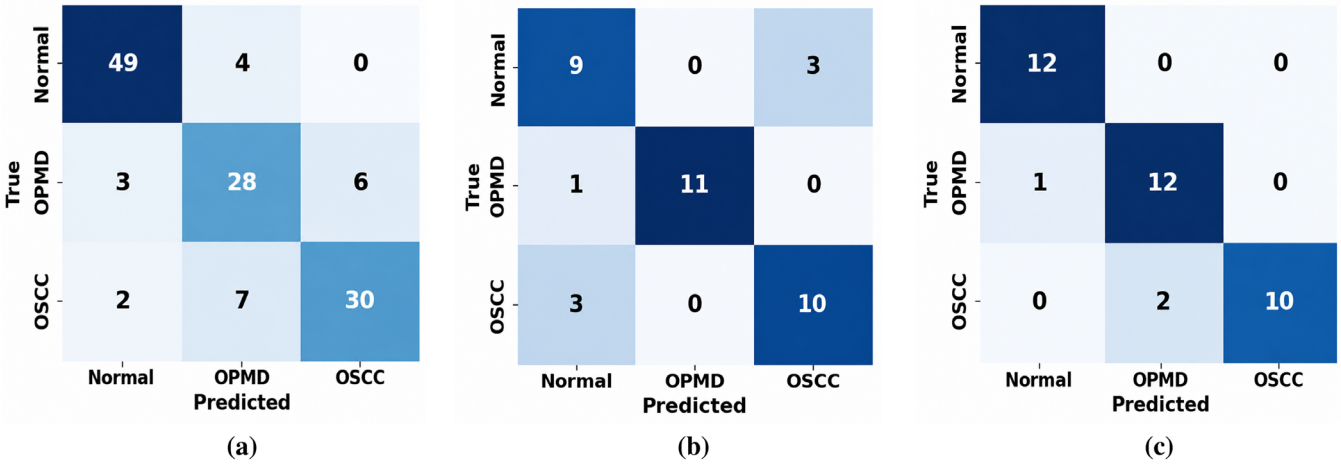


FIGURE 7 | Confusion matrices for different input modalities: (a) only images, (b) only spectra, and (c) combined images + spectra.

TABLE 2 | Performance metrics based on the confusion matrix (only images).

Class	Pre	Sen	F1	Spec	Acc
Normal	90.74	92.45	91.59	93.42	93.02
OPMD	71.79	75.68	73.68	88.04	84.50
OSCC	83.33	76.92	80.00	93.33	88.37
Average	81.96	81.68	81.76	91.60	88.63

TABLE 3 | Performance metrics based on the confusion matrix (only spectra).

Class	Pre	Sen	F1	Spec	Acc
Normal	69.23	75.00	72.00	84.00	81.08
OPMD	100.00	91.67	95.65	100.00	97.30
OSCC	76.92	76.92	76.92	87.50	83.78
Average	82.05	81.20	81.53	90.50	87.39

TABLE 4 | Performance metrics based on the confusion matrix (both: images + spectra).

Class	Pre	Sen	F1	Spec	Acc
Normal	92.31	100.00	96.00	96.00	97.23
OPMD	85.71	92.31	88.89	91.67	91.89
OSCC	100.00	83.33	90.91	100.00	94.59
Average	92.67	91.88	91.93	95.89	94.59

assessment at clinically relevant sites [32]. Embedding both modalities in a single device enhances the effectiveness of oral cancer diagnosis.

In addition to diagnostic accuracy, the device offers key advantages—it is portable, label-free, noninvasive, and cost-effective—making it competitive with currently available commercial

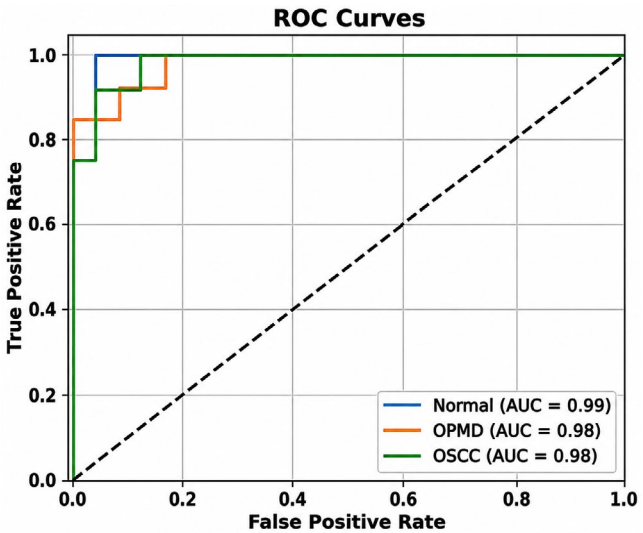


FIGURE 8 | ROC-AUC curve.

platforms. The bimodal design is particularly impactful because it leverages complementary strengths: imaging provides spatial fluorophore mapping and biopsy guidance, while spectroscopy captures biochemical sensitivity. This underscores the need to adopt bimodal approaches for reliable diagnosis using fluorescence-based optical techniques. Furthermore, the integration of AI/ML enables real-time classification and decision support, a critical feature for mass screening and effective healthcare delivery in resource-limited settings.

Table 5 summarizes the performance of existing point-of-care devices. Commercial systems such as VELscope, Identafi, OralID, and ViziLite PRO primarily rely on autofluorescence or multi-spectral imaging. While these devices are portable and user-friendly, their diagnostic performance remains modest, with sensitivities of 70%–77% and specificities as low as 27%–66%. High false-positive rates, subjective interpretation, and lack of biopsy guidance further limit their effectiveness in detecting early disease.

More advanced systems like OralScan, which integrates tissue fluorescence and diffuse reflectance with AI analysis,

TABLE 5 | Comparison of point-of-care devices used for oral precancer and cancer diagnosis.

Device/ screening tool	Technique/ working principle	Portable	AI/ML (purpose)	User friendly	Biopsy guidance	Cost	Label- free imaging	Sensitivity/ specificity	Key advantages	Limitations	Market status
VELscope (LED Dental)	Autofluorescence	Yes	No	Easy	No	High	Yes	76%/66.3% [50]	Noninvasive, real-time visualization	High false positives, subjective interpretation	Commercialized
Identafi (StarDental)	Multispectral fluorescence and reflectance	Yes	No	Moderate expertise	No	High	Yes	73%/46.2% [51]	Multi- wavelength detection	Requires training, variable specificity	Commercialized
OralID (Forward Science)	Autofluorescence	Yes	No	Easy	No	Medium	Yes	74%/89% [52]	Portable, easy to use	Limited depth penetration	Commercialized
ViziLite (DenMat)	Blue-light based fluorescence imaging (requires acetic acid)	Yes	No	Easy	No	Low	No	77.3%/27.8% [53]	Simple, low-cost	Low specificity, adjunctive only	Commercialized
OralScan (Sascan)	Hand-held multispectral imaging camera using tissue fluorescence and diffuse reflectance with cloud-based AI analysis	Yes	Yes (classification and detection)	Moderate expertise	Yes	Medium	Yes	82%/96.6% [54]	ML-assisted, real-time detection, biopsy guidance	Needs large- scale clinical adoption	Commercialized
SBBD (Smartphone- based bimodal device)	Fluorescence spectroscopy and imaging with AI- based classification	Yes	Yes (classification)	Easy	Yes (using R/G imaging)	Medium	Yes	91.88%/95.89%	High accuracy, AI-enhanced, real-time diagnosis	Needs large- scale clinical validation	Prototype/ preclinical stage

show improved diagnostic performance (82% sensitivity and 96.6% specificity) and provide biopsy guidance. However, these systems have not yet achieved widespread clinical adoption.

In contrast, the proposed SBBD demonstrates clear advantages. By combining fluorescence spectroscopy with imaging and AI-driven classification, it achieves higher diagnostic accuracy (91.9% sensitivity and 95.9% specificity) than most commercial devices. The IM offers lesion mapping and biopsy guidance, while spectroscopy provides biochemical insights that imaging alone cannot capture. This integration directly addresses the limitations of both modalities, making the bimodal design a compelling requirement for next-generation devices.

Beyond diagnostic performance, the SBBD's portability, low cost, and real-time operation make it particularly well-suited for large-scale deployment. Unlike many commercial devices that are expensive or require trained operators, this system is designed for accessibility in community and low-resource healthcare settings. Compared with conventional oral autofluorescence imaging systems, the SBBD provides a compact smartphone-based bimodal platform capable of performing both fluorescence imaging and spectroscopy within a single handheld device. The integration of these complementary modalities enhances lesion assessment by combining spatial fluorescence visualization with quantitative spectral analysis, thereby improving diagnostic confidence. Furthermore, the system enables label-free and noninvasive evaluation of oral lesions. As the device was specifically developed for in vivo oral cancer detection, clinical usability, safety, sterility, and ease of operation were integral considerations in its design. To minimize the risk of cross-contamination, a single-use biocompatible disposable tube fabricated from medical-grade PLA was incorporated into the system. In addition, all externally exposed device surfaces were disinfected with medical-grade alcohol before each clinical examination. The device utilizes a 405 nm excitation laser operating in the visible wavelength range. The tissue exposure time is limited to less than 2 min, ensuring safe clinical operation while minimizing patient discomfort and maintaining compliance with safety considerations. The smartphone-integrated interface facilitates rapid data acquisition and user-friendly operation during routine clinical examinations. Although the preliminary clinical results demonstrate promising diagnostic performance, further validation across larger multicenter cohorts and a broader range of pathological conditions will be necessary to establish the clinical robustness, generalizability, and translational potential of the proposed system.

Taken together, the comparative analysis underscores the limitations of existing single-modality devices and highlights the necessity of combining complementary techniques. By integrating spectroscopy, imaging, and AI within a single portable platform, the proposed SBBD enhances diagnostic accuracy, enables biopsy guidance, and supports population-level screening. These features strongly position it as a promising candidate for clinical translation and commercialization, with the potential to transform early detection and management of oral precancer and cancer.

4 | Conclusion

This study demonstrates the feasibility of using a portable, smartphone-based bimodal fluorescence imaging and spectroscopy device for the early detection and classification of oral cancer. By integrating spectral and imaging features within a unified classification framework, the system achieved high diagnostic performance, with macro-average accuracy exceeding 94%. These findings highlight the complementary role of biochemical and morphological information in distinguishing normal, precancerous, and cancerous tissue, underscoring the clinical potential of this approach for large-scale screening and real-time biopsy guidance. Future studies on larger and more diverse cohorts will further validate its translational impact in improving oral cancer screening, particularly in resource-limited settings.

Acknowledgments

Nemichand acknowledges the Institute Fellowship provided by the Indian Institute of Technology Kanpur. The authors acknowledge Dr. Amar Nath Sah for his assistance in developing the initial prototype of the device. The authors also thank Mrs. Subrata Mishra and Ms. Bishakha Dash for their assistance in clinical data collection for this study.

Funding

This study was funded by PhotoSpIMeDx Pvt. Ltd., a company incubated at IIT Kanpur and SOA University, Odisha.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

1. H. Sung, J. Ferlay, R. L. Siegel, et al., "Global Cancer Statistics 2020: Globocan Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries," *CA: A Cancer Journal for Clinicians* 71 (2021): 209–249.
2. E. P. Tranby, L. J. Heaton, S. L. Tomar, et al., "Oral Cancer Prevalence, Mortality, and Costs in Medicaid and Commercial Insurance Claims Data," *Cancer Epidemiology, Biomarkers & Prevention* 31 (2022): 1849–1857.
3. S. Warnakulasuriya and A. Kerr, "Oral Cancer Screening: Past, Present, and Future," *Journal of Dental Research* 100 (2021): 1313–1320.
4. A. Chaurasia, S. I. Alam, and N. Singh, "Oral Cancer Diagnostics: An Overview," *National Journal of Maxillofacial Surgery* 12 (2021): 324–332.
5. S. Hussain, I. Mubeen, N. Ullah, et al., "Modern Diagnostic Imaging Technique Applications and Risk Factors in the Medical Field: A Review," *BioMed Research International* 2022 (2022): 5164970.
6. D. Shin, N. Vigneswaran, A. Gillenwater, and R. Richards-Kortum, "Advances in Fluorescence Imaging Techniques to Detect Oral Cancer and Its Precursors," *Future Oncology* 6 (2010): 1143–1154.

7. K. Yadav, I. Cree, A. Field, et al., "Importance of Cytopathologic Diagnosis in Early Cancer Diagnosis in Resourceconstrained Countries," *JCO Global Oncology* 8 (2022): e2100337.
8. M. S. Rahman, N. Ingle, D. Roblyer, et al., "Evaluation of a Low-Cost, Portable Imaging System for Early Detection of Oral Cancer," *Head & Neck Oncology* 2 (2010): 10.
9. B. L. Ziober, M. G. Mauk, E. M. Falls, et al., "Lab-on-a-Chip for Oral Cancer Screening and Diagnosis," *Head & Neck* 30 (2008): 111–121.
10. H. G. Dailah, "Mobile Health (Mhealth) Technology in Early Detection and Diagnosis of Oral Cancer—A Scoping Review of the Current Scenario and Feasibility," *Journal of Healthcare Engineering* 2022 (2022): 4383303.
11. D. Pandya, A. K. Nagarajappa, S. Reddy, and M. Bhasin, "Lab-On-a-Chip-Oral Cancer Diagnosis at Your Door Step," *Journal of International Oral Health* 7 (2015): 122.
12. S. R. Bisht, P. Mishra, D. Yadav, R. Rawal, and K. P. Mercado-Shekhar, "Current and Emerging Techniques for Oral Cancer Screening and Diagnosis: A Review," *Progress in Biomedical Engineering* 3 (2021): 042003.
13. M. O. Shaikh, C.-M. Lin, D.-H. Lee, W. F. Chiang, I. H. Chen, and C. H. Chuang, "Portable Pen-Like Device With Miniaturized Tactile Sensor for Quantitative Tissue Palpation in Oral Cancer Screening," *IEEE Sensors Journal* 20 (2020): 9610–9617.
14. Y. Wang, M. Raj, H. S. McGuff, et al., "Portable Oral Cancer Detection Using a Miniature Confocal Imaging Probe With a Large Field of View," *Journal of Micromechanics and Microengineering* 22 (2012): 065001.
15. S. Rajendran, A. Porwal, K. Anjali, et al., "Portable Iot Devices in Healthcare for Health Monitoring and Diagnostics," in *Internet Things Bioelectronics: Emerging Technologies Applications*, ed. H. Murthy, M. Zurek-Mortka, V. Jha Pillai, and K. Pradeep Kumar (Wiley, 2024), 263–296.
16. M. Zarei, "Portable Biosensing Devices for Point-of-Care Diagnostics: Recent Developments and Applications," *TrAC Trends in Analytical Chemistry* 91 (2017): 26–41.
17. G. S. Reddy, K. E. Rao, K. K. Kumar, P. C. Sekhar, K. L. Prakash Chandra, and B. V. Ramana Reddy, "Diagnosis of Oral Cancer: The Past and Present," *Journal of Orofacial Sciences* 6 (2014): 10–16.
18. C. Scully, J. V. Bagan, C. Hopper, and J. B. Epstein, "Oral Cancer: Current and Future Diagnostic Techniques," *American Journal of Dentistry* 21 (2008): 199–209.
19. D. A. Andreadis, A.-M. Pavlou, and P. Panta, "Biopsy and Oral Squamous Cell Carcinoma Histopathology," in *Oral Cancer Detection: Novel Strategies and Clinical Impact*, ed. P. Panta (Springer, 2018), 133–151.
20. S. Chen, M. Forman, P. M. Sadow, and M. August, "The Diagnostic Accuracy of Incisional Biopsy in the Oral Cavity," *Journal of Oral and Maxillofacial Surgery* 74 (2016): 959–964.
21. G. Yang, L. Wei, B. K. Thong, et al., "A Systematic Review of Oral Biopsies, Sample Types, and Detection Techniques Applied in Relation to Oral Cancer Detection," *Biotech* 11 (2022): 5.
22. S. Shukla, B. S. Deo, C. Vishwakarma, et al., "A Smartphone-Based Standalone Fluorescence Spectroscopy Tool for Cervical Precancer Diagnosis in Clinical Conditions," *Journal of Biophotonics* 17 (2024): e202300468.
23. H.-C. Wang, M.-T. Tsai, and C.-P. Chiang, "Visual Perception Enhancement for Detection of Cancerous Oral Tissue by Multi-Spectral Imaging," *Journal of Optics* 15 (2013): 055301.
24. N. Subhash, J. Mallia, S. S. Thomas, et al., "Oral Cancer Detection Using Diffuse Reflectance Spectral Ratio r540/r575 of Oxygenated Hemoglobin Bands," *Journal of Biomedical Optics* 11 (2006): 014018.
25. B. H. Malik, J. M. Jabbour, S. Cheng, et al., "A Novel Multimodal Optical Imaging System for Early Detection of Oral Cancer," *Oral Surgery, Oral Medicine, Oral Pathology and Oral Radiology* 121 (2016): 290–300.
26. J. L. Norris and R. M. Caprioli, "Analysis of Tissue Specimens by Matrix-Assisted Laser Desorption/Ionization Imaging Mass Spectrometry in Biological and Clinical Research," *Chemical Reviews* 113 (2013): 2309–2342.
27. N. Chacko and R. Ankri, "Non-Invasive Early-Stage Cancer Detection: Current Methods and Future Perspectives," *Clinical and Experimental Medicine* 25 (2024): 17.
28. C. Balas, "Review of Biomedical Optical Imaging—A Powerful, Non-Invasive, Non-Ionizing Technology for Improving In Vivo Diagnosis," *Measurement Science and Technology* 20 (2009): 104020.
29. U. M. Pal, M. Saxena, G. A. Vishnu, et al., "Optical Spectroscopy-Based Imaging Techniques for the Diagnosis of Breast Cancer: A Novel Approach," *Applied Spectroscopy Reviews* 55 (2020): 778–804.
30. P. Kumar, S. Rathod, and A. Pradhan, "Detection of Oral Mucosal Lesions by the Fluorescence Spectroscopy and Classification of Cancerous Stages by Support Vector Machine," *Lasers in Medical Science* 39 (2024): 42.
31. A. N. Sah, P. Kumar, and A. Pradhan, "In-Vivo Testing of Oral Mucosal Lesions With an In-House Developed Portable Imaging Device and Comparison With Spectroscopy Results," *Journal of Fluorescence* 33 (2023): 1375–1383.
32. Nemichand, S. Shukla, B. S. Deo, et al., "Smartphone-Based Bi-modal Device (SBBDD) for Oral Precancer Diagnosis and Biopsy Guidance in Clinical Settings," *Optics Letters* 50 (2025): 1993–1996.
33. C. Krafft, M. Schmitt, I. W. Schie, et al., "Label-Free Molecular Imaging of Biological Cells and Tissues by Linear and Nonlinear Raman Spectroscopic Approaches," *Angewandte Chemie International Edition* 56 (2017): 4392–4430.
34. M. Sauer, J. Hofkens, and J. Enderlein, *Handbook of Fluorescence Spectroscopy and Imaging: From Ensemble to Single Molecules* (John Wiley & Sons, 2010).
35. A. C. Croce and G. Bottiroli, "Autofluorescence Spectroscopy and Imaging: A Tool for Biomedical Research and Diagnosis," *European Journal of Histochemistry: EJH* 58 (2014): 2461.
36. J. A. Kim, D. J. Wales, and G.-Z. Yang, "Optical Spectroscopy for In Vivo Medical Diagnosis—A Review of the State of the Art and Future Perspectives," *Progress in Biomedical Engineering* 2 (2020): 042001.
37. S. Maryam, M. S. Nogueira, R. Gautam, et al., "Label-Free Optical Spectroscopy for Early Detection of Oral Cancer," *Diagnostics* 12 (2022): 2896.
38. P. Thapa, V. Singh, S. Bhatt, et al., "Multimodal Fluorescence Imaging and Spectroscopic Techniques for Oral Cancer Screening: A Real-Time Approach," *Methods and Applications in Fluorescence* 11 (2023): 045008.
39. M. G. Müller, T. A. Valdez, I. Georgakoudi, et al., "Spectroscopic Detection and Evaluation of Morphologic and Biochemical Changes in Early Human Oral Carcinoma," *Cancer: An Interdisciplinary, International Journal of the American Cancer Society* 97 (2003): 1681–1692.
40. S. Sivabalan, C. P. Vedeswari, S. Jayachandran, et al., "In Vivo Native Fluorescence Spectroscopy and Nicotinamide Adenine Dinucleotide/Flavin Adenine Dinucleotide Reduction and Oxidation States of Oral Submucous Fibrosis for Chemopreventive Drug Monitoring," *Journal of Biomedical Optics* 15 (2010): 017010.
41. F. R. de Oliveira Silva, M. H. Bellini, V. R. Tristão, et al., "Intrinsic Fluorescence of Protoporphyrin IX From Blood Samples Can Yield Information on the Growth of Prostate Tumours," *Journal of Fluorescence* 20 (2010): 1159–1165.
42. K. Onizawa, H. Saginoya, Y. Furuya, and H. Yoshida, "Fluorescence Photography as a Diagnostic Method for Oral Cancer," *Cancer Letters* 108 (1996): 61–66.

43. P. M. Lane, T. Gilhuly, P. Whitehead, et al., "Simple Device for the Direct Visualization of Oral-Cavity Tissue Fluorescence," *Journal of Biomedical Optics* 11 (2006): 024006.
44. M. Inaguma and K. Hashimoto, "Porphyrin-Like Fluorescence in Oral Cancer: In Vivo Fluorescence Spectral Characterization of Lesions by Use of a Near-Ultraviolet Excited Autofluorescence Diagnosis System and Separation of Fluorescent Extracts by Capillary Electrophoresis," *Cancer: An Interdisciplinary, International Journal of the American Cancer Society* 86 (1999): 2201–2211.
45. C. F. Poh, S. Ng, K. W. Berean, et al., "Biopsy and Histopathologic Diagnosis of Oral Premalignant and Malignant Lesions," *Journal of the Canadian Dental Association* 74 (2008): 283–288.
46. L.-J. Tseng, A. Matsuyama, and V. MacDonald-Dickinson, "Histology: The Gold Standard for Diagnosis?," *Canadian Veterinary Journal* 64 (2023): 389.
47. B. S. Deo, M. Pal, P. K. Panigrahi, and A. Pradhan, "An Ensemble Deep Learning Model With Empirical Wavelet Transform Feature for Oral Cancer Histopathological Image Classification," *International Journal of Data Science and Analytics* 20 (2024): 1–18.
48. S. Dixit, A. Kumar, and K. Srinivasan, "A Current Review of Machine Learning and Deep Learning Models in Oral Cancer Diagnosis: Recent Technologies, Open Challenges, and Future Research Directions," *Diagnostics* 13 (2023): 1353.
49. S. Shukla, A. N. Sah, D. Hatiboruah, S. Ahirwar, P. Nath, and A. Pradhan, "Design, Fabrication and Testing of 3d Printed Smartphone-Based Device for Collection of Intrinsic Fluorescence From Human Cervix," *Scientific Reports* 12 (2022): 11192.
50. R. S. Ganga, D. Gundre, S. Bansal, P. M. Shirsat, P. Prasad, and R. S. Desai, "Evaluation of the Diagnostic Efficacy and Spectrum of Autofluorescence of Benign, Dysplastic and Malignant Lesions of the Oral Cavity Using Velscope," *Oral Oncology* 75 (2017): 67–74.
51. D. Sharma, J. Rimal, I. K. Maharjan, et al., "Evaluation of Oral Potentially Malignant Disorders With Autofluorescence, Reflectance Spectroscopy and Vital Staining and Their Correlation With Histopathology—Hospital Based Prospective Study," *Oral Oncology* 118 (2021): 105312.
52. R. Saini, S. Cantore, S. R. Saini, et al., "Efficacy of Fluorescence Technology vs Conventional Oral Examination for the Early Detection of Oral Pre-Malignant Lesions. A Clinical Comparative Study," *Endocrine, Metabolic & Immune Disorders—Drug Targets* 19 (2019): 852–858.
53. T. Sambandham, K. Masthan, M. S. Kumar, and A. Jha, "The Application of Vizilite in Oral Cancer," *Journal of Clinical and Diagnostic Research* 7 (2013): 185–186.
54. N. Subhash, S. Anand, R. Prasanna, et al., "Bimodal Multispectral Imaging System With Cloud-Based Machine Learning Algorithm for Real-Time Screening and Detection of Oral Potentially Malignant Lesions and Biopsy Guidance," *Journal of Biomedical Optics* 26 (2021): 086003.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Figure S1:** Optical schematic of smartphone-based bimodal device (SBBD): (a) imaging module and (b) spectroscopy module (CL: collimating lens; BS: beam splitter; LPF: long-pass filter; TG: transmission grating; FL: focusing lens; and SD: smartphone detector). **Figure S2:** CAD design of the SBBD device. (a) Assembly of the device components, (b) mobile holder, (c) grating holder, (d) optical unit, and (e) device stand. **Figure S3:** Pixel-wavelength calibration curve for spectroscopy module. **Figure S4:** Fluorescence spectra of liquid: (a) FAD and (b) PPIX fluorophore at varying concentrations, captured using the SBBD. **Figure S5:** Calibration for fluorescence imaging: (a) Fluorescence images of FAD liquid phantom at concentrations of 62.6, 31.2, 15.6, 3.9, 1.9, and 1.0 μM (right to left). (b) Fluorescence images of PPIX liquid phantom at concentrations of 100, 25, 6.250, 1.56, 0.39, and 0.20 μM .